Introduction to Linear Regression: Frequentist, Bayesian and Other Perspectives

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1 Linear Models

In supervised learning, we observe data of the form

$$(X_1, Y_1), \ldots, (X_n, Y_n),$$

where each X_i denotes a vector of predictors and Y_i denotes a response.

The goal is to use the observed data to construct a prediction rule

$$\hat{f}(x),$$

which predicts the response Y when the predictors take the value x.

Examples include predicting house prices from characteristics of houses, predicting sales from advertising budgets, predicting whether an email is spam, and predicting a medical outcome from patient measurements.

Linear regression is one of the simplest and most important supervised learning methods. It assumes that the dependence of the response on the predictors is approximately linear.

A useful general model for regression is

$$Y = f^*(X) + \varepsilon,$$

where f^* is the true regression function and ε is noise.

In linear regression, we approximate f^* by a linear function. For example, with predictors

$$X = (X_1, \ldots, X_p),$$

we approximate f^* by

$$\eta(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

Real regression functions (do they exist?) are rarely exactly linear. Nevertheless, linear regression is extremely useful both conceptually and practically and forms the basic building block of more complex models.

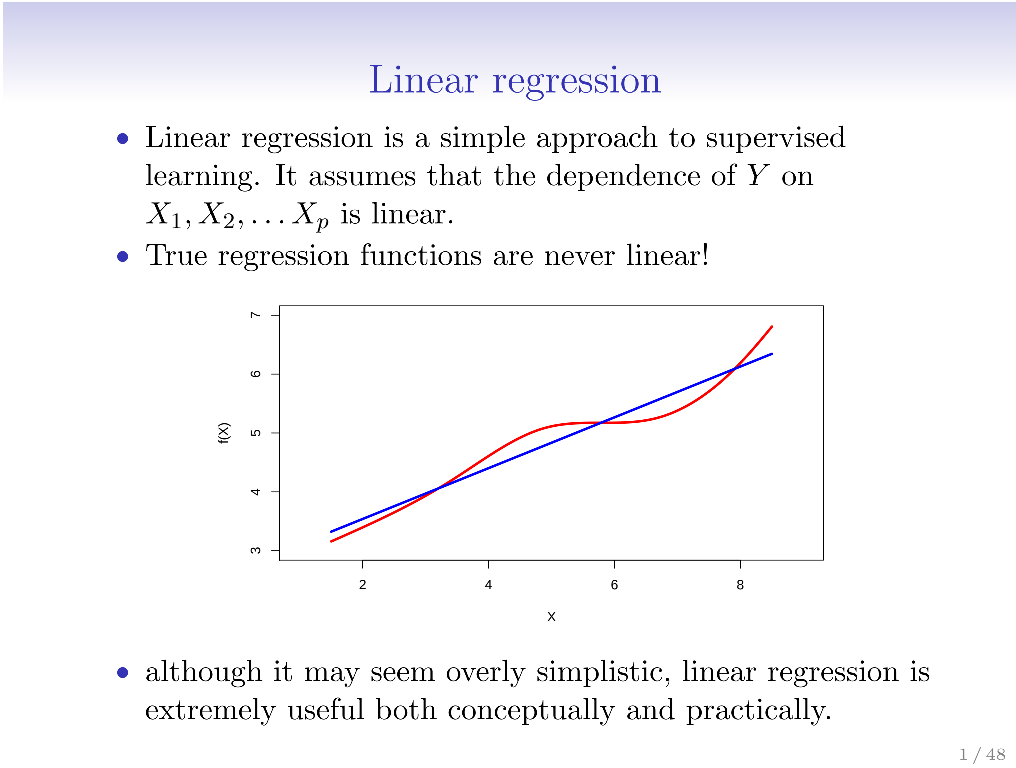

Figure 1: Linear regression should usually be viewed as an approximation. The true relationship may be nonlinear, but a linear model may still capture an important first-order trend.

A useful quote, due to George Box, is:

"All models are wrong, but some are useful."

We do not necessarily believe that the true relationship is exactly linear. Rather, we use linear models because they are simple, interpretable, computationally convenient, and often surprisingly effective.

1.1 Simple Linear Regression

We first consider the case of a single predictor. The simple linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Here β_0 is the intercept and β_1 is the slope. The error terms are often modeled as i.i.d zero mean random variables. Sometimes it may be reasonable to assume

$$\varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The advertising data provide a useful motivating example. We may ask whether sales are related to advertising expenditure on TV, radio, and newspapers.

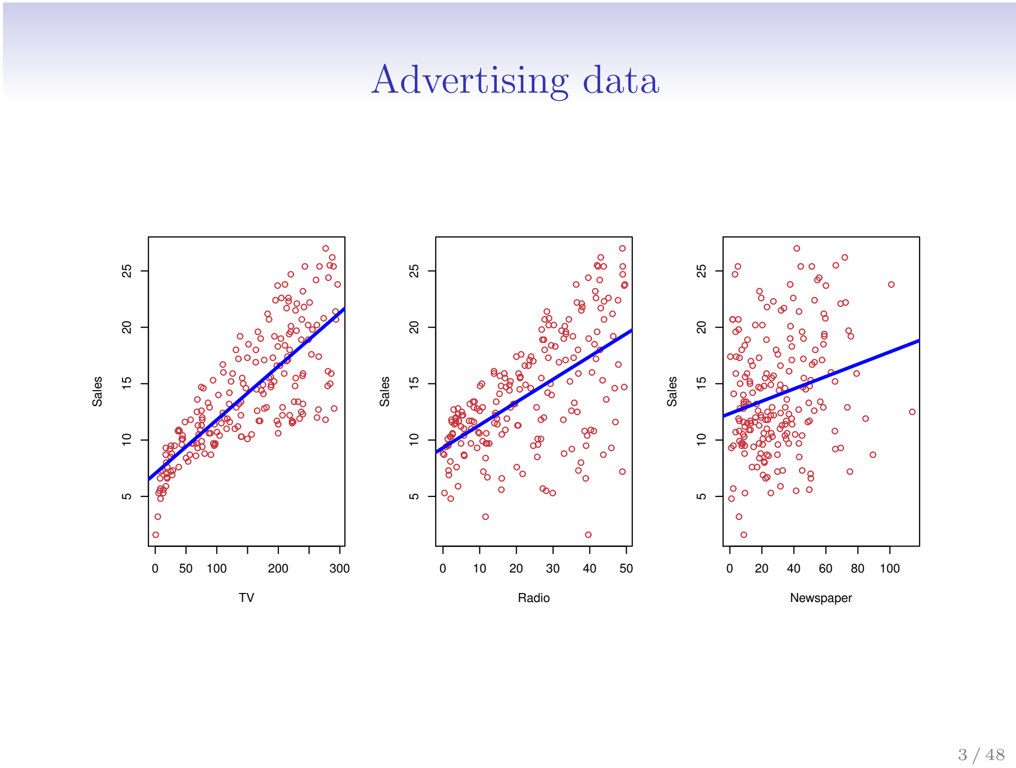

Figure 2: Advertising data: scatterplots of sales versus TV, radio, and newspaper advertising budgets. Visualizing the data is often the first step before fitting a model.

For a fitted line, the fitted value for observation i is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

The residual is

$$e_i = y_i - \hat{y}_i.$$

The residual sum of squares is

$$\text{RSS} = \sum_{i=1}^{n} e_i^2.$$

Equivalently,

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

Least squares chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize this residual sum of squares.

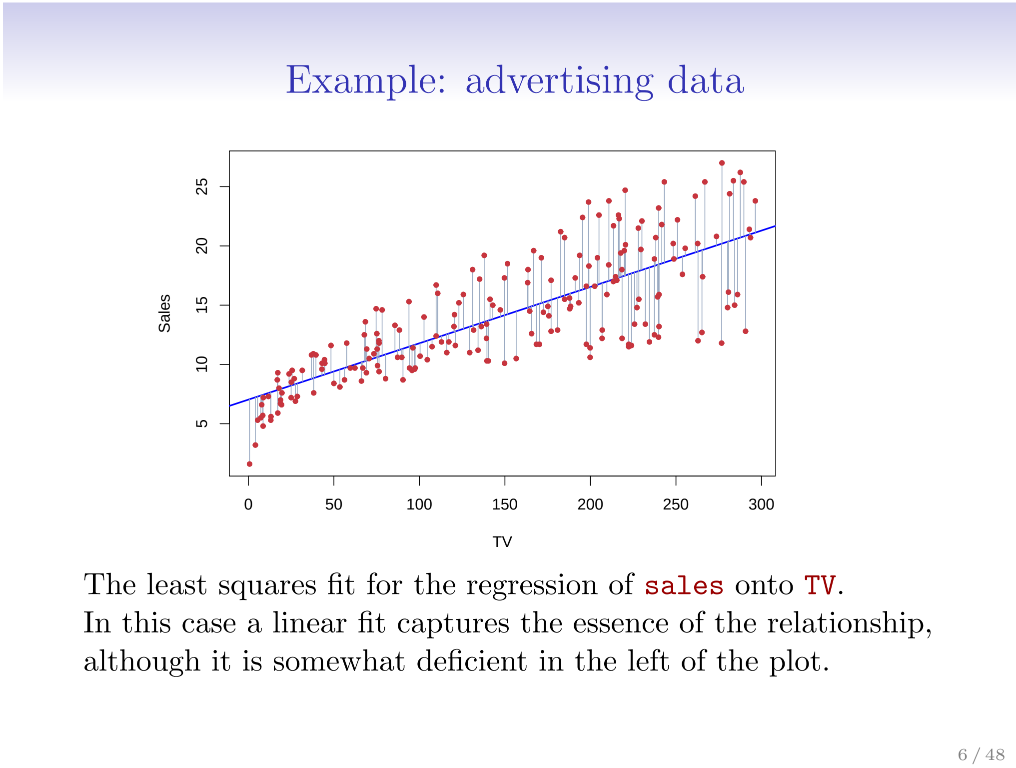

Figure 3: Least squares fit for sales on TV advertising. The vertical segments represent residuals. The least squares line minimizes the sum of squared residuals.

In simple linear regression, the least squares estimators have explicit formulas:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

In the next section, we will see how to derive these explicit formulas.

Exercise: What if we use the absolute value loss instead of the squared loss? Can we make some comparions of the two methods?

Exercise: What is the line we get if we change the notion of distance of a point from a line, i.e, from vertical to perpendicular?

1.2 Multiple Linear Regression

We now have several predictors. Suppose that for each observation i, we observe predictors

$$x_{i1}, \ldots, x_{im}.$$

The multiple linear regression model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_m x_{im} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Equivalently,

$$y_i = f^*(x_i) + \varepsilon_i,$$

where

$$f^*(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m.$$

In the advertising example, a multiple regression model is

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot \text{newspaper} + \varepsilon.$$

For two predictors, the fitted regression function is a plane.

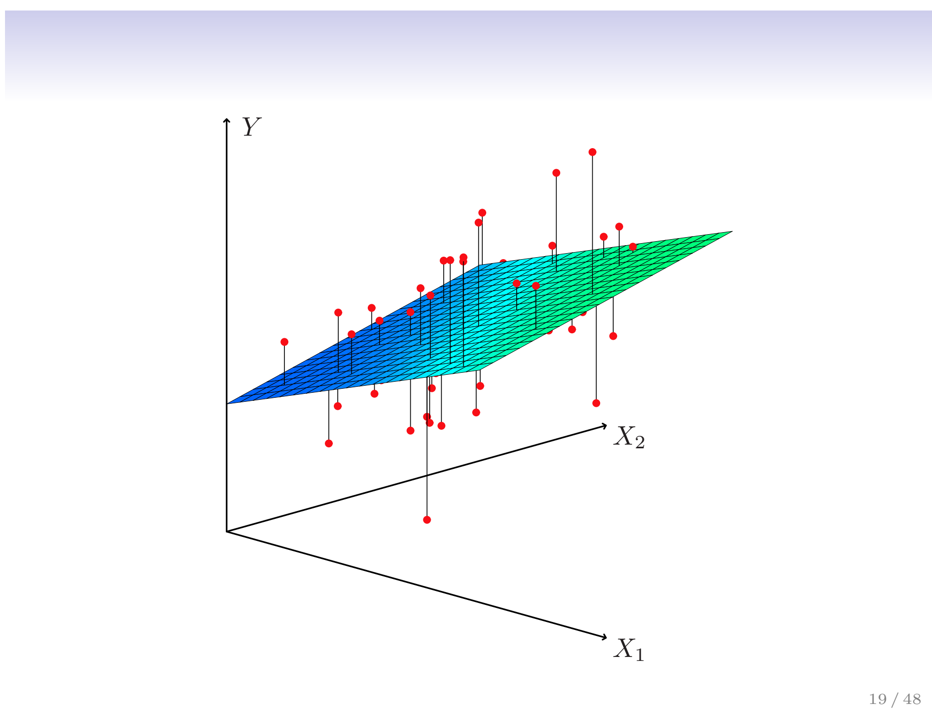

Figure 4: With two predictors, the fitted regression function is a plane. Least squares chooses the plane that minimizes the sum of squared vertical distances.

1.2.1 Matrix Form

Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix},$$

and

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1m} \\ 1 & x_{21} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nm} \end{pmatrix}.$$

Then the model is

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I_n).$$

The unknown parameters are

$$\beta_0, \ldots, \beta_m, \sigma.$$

1.3 Least Squares Estimation

The least squares estimator minimizes

$$S(\beta) = \|y - X\beta\|^2.$$

Expanding,

$$S(\beta) = (y - X\beta)^\top (y - X\beta).$$

Therefore

$$S(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta.$$

Differentiating with respect to β,

$$\nabla S(\beta) = -2X^\top y + 2X^\top X\beta.$$

At the minimizer $\hat{\beta}$, the gradient is zero:

$$-2X^\top y + 2X^\top X\hat{\beta} = 0.$$

Thus

$$X^\top X\hat{\beta} = X^\top y.$$

Assuming $X^\top X$ is invertible (true if X has full column rank),

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

This is the explicit closed form expression for the ordinary least squares estimator.

Exercise: What if X is not full column rank?

The fitted values are

$$\hat{y} = X\hat{\beta}.$$

The vector $\hat{y}$ is the orthogonal projection of y onto the column space of X. This means that the residual vector

$$y - \hat{y}$$

is orthogonal to every column of X. Algebraically,

$$X^\top (y - X\hat{\beta}) = 0.$$

A key identity is the Pythoagorean identity

$$S(\beta) = S(\hat{\beta}) + \|X\hat{\beta} - X\beta\|^2. \tag{1}$$

1.4 Interpreting Regression Coefficients

In multiple linear regression, the coefficient β_j is commonly interpreted as the expected change in the response for a one-unit increase in X_j, holding all other predictors fixed. The usual interpretation is useful, but it should be applied with care.

- This may make sense if the linear model is correct. What if the linear model is not correct?
- Typically the predictors will be correlated, like in the ad data, the TV, radio and newspaper ad expenditures are likely to be correlated. Here, if we increase one predictor, the other predictor will tend to increase as well. In this case, we do not really have enough data to estimate the expected change in a predictor keeping other predictors fixed.
- We probably can think of the coefficient β_j as the expected change in the response for a one-unit increase in X_j, keeping other predictors fixed and *adjusted for the presence of other predictors in the model.*

Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

Correlations:

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

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Figure 5: Multiple regression output for the advertising data. Notice that newspaper has a weak coefficient after adjusting for TV and radio, even though it may have some marginal relationship with sales.

A more fundamental way to understand regression coefficients is through the idea of the best linear predictor. Suppose that

$$(Y, X_1, \dots, X_p)$$

has some joint distribution. We do not need to assume that the true conditional expectation of Y given $X_1, \dots, X_p$ is linear. We can still ask for the best linear approximation to Y based on the predictors.

Let

$$X = (X_1, \dots, X_p)^T.$$

The best linear predictor is defined as the solution of

$$(\beta_0^*, \beta^*) = \arg\min_{\beta_0, \beta} \mathbb{E}\left[\left\{Y - \beta_0 - \beta^T X\right\}^2\right].$$

The normal equations (take derivatives and set to 0) imply

$$\mathbb{E}\left[Y - \beta_0^* - (\beta^*)^T X\right] = 0$$

and

$$\mathbb{E}\left[X\left\{Y-\beta_0^*-(\beta^*)^TX\right\}\right]=0.$$

Thus the residual from the best linear predictor is uncorrelated with each predictor.

If we center all variables, so that

$$\mathbb{E}Y=0, \qquad \mathbb{E}X_j=0,$$

then $\beta_0^*=0$, and the problem becomes

$$\beta^*=\arg\min_\beta \mathbb{E}\left[\left(Y-\beta^TX\right)^2\right].$$

The normal equations are

$$\mathbb{E}\left[X(Y-\beta^TX)\right]=0.$$

Equivalently,

$$\mathbb{E}(XX^T)\beta=\mathbb{E}(XY).$$

Writing

$$\Sigma_{XX}=\mathbb{E}(XX^T), \qquad \Sigma_{XY}=\mathbb{E}(XY),$$

we obtain

$$\beta^*=\Sigma_{XX}^{-1}\Sigma_{XY}.$$

This formula gives the more general population version of least squares. The ordinary least squares estimator computed from data is a special case of this formula (why?).

There is also an important interpretation of the individual coefficient β_j^*. Let X_{-j} denote all predictors except X_j. First regress X_j linearly on the other predictors, and define the residual

$$R_j=X_j-\Pi(X_j\mid X_{-j}),$$

where $\Pi(X_j\mid X_{-j})$ denotes the best linear predictor of X_j from X_{-j}. Thus R_j is the part of X_j that cannot be linearly explained by the other predictors.

Then the coefficient of X_j in the full multiple regression is

$$\beta_j^*=\frac{\mathbb{E}(R_jY)}{\mathbb{E}(R_j^2)}.$$

Equivalently, if we also regress Y linearly on X_{-j} and let

$$R_Y=Y-\Pi(Y\mid X_{-j}),$$

then

$$\beta_j^*=\frac{\mathbb{E}(R_jR_Y)}{\mathbb{E}(R_j^2)}.$$

This is the precise meaning of “adjusting for the other variables.” The coefficient β_j^* is the same as the simple linear regression coefficient between the parts of Y and X_j left over after removing its linear dependence on the other predictors.

Exercise: Show the above interpretation holds.

Thus β_j^* is not simply the marginal relationship between Y and X_j. It is the marginal relationship between Y and the residualized version of X_j, after regressing out the other predictors. It is an adjusted coefficient after regressing out the other predictors.

1.5 Qualitative Predictors

Not all predictors are numerical. Some predictors are categorical or qualitative.

For example, suppose we want to compare average credit card balance between males and females. We can define a dummy variable

$$x_i = \begin{cases} 1, & \text{if person } i \text{ is female,} \\ 0, & \text{if person } i \text{ is male.} \end{cases}$$

Then the model becomes

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Equivalently,

$$y_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i, & \text{if person } i \text{ is female,} \\ \beta_0 + \varepsilon_i, & \text{if person } i \text{ is male.} \end{cases}$$

Thus β_0 is the mean response for the baseline group, and β_1 is the difference between the two groups.

For a categorical variable with more than two levels, we use multiple dummy variables. If a variable has K levels, we have to use

$$K - 1$$

dummy variables. The omitted level is called the baseline level.

For example, if ethnicity has levels Asian, Caucasian, and African American, we may define

$$x_{i1} = \begin{cases} 1, & \text{if person } i \text{ is Asian,} \\ 0, & \text{otherwise,} \end{cases}$$

and

$$x_{i2} = \begin{cases} 1, & \text{if person } i \text{ is Caucasian,} \\ 0, & \text{otherwise.} \end{cases}$$

Then African American is the baseline group.

1.6 Nonlinear Function Fitting

1.6.1 Polynomial Regression

The linear regression model assumes that the response depends linearly on the predictors. *One of the great strengths of linear regression is that we are free to choose predictors and nonlinear functions of predictors in the model.* The model only needs to be linear in the unknown coefficients.

For example, suppose that the relationship between a response Y and a predictor X is nonlinear. We can introduce transformed predictors and fit a polynomial model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon.$$

This is still a linear regression model because it is linear in the unknown coefficients

$$\beta_0, \beta_1, \ldots, \beta_d.$$

The predictors have simply been transformed into

$$X, X^2, \ldots, X^d.$$

For example, a quadratic model for the Auto data is

$$\text{mpg} = \beta_0 + \beta_1 \cdot \text{horsepower} + \beta_2 \cdot \text{horsepower}^2 + \varepsilon.$$

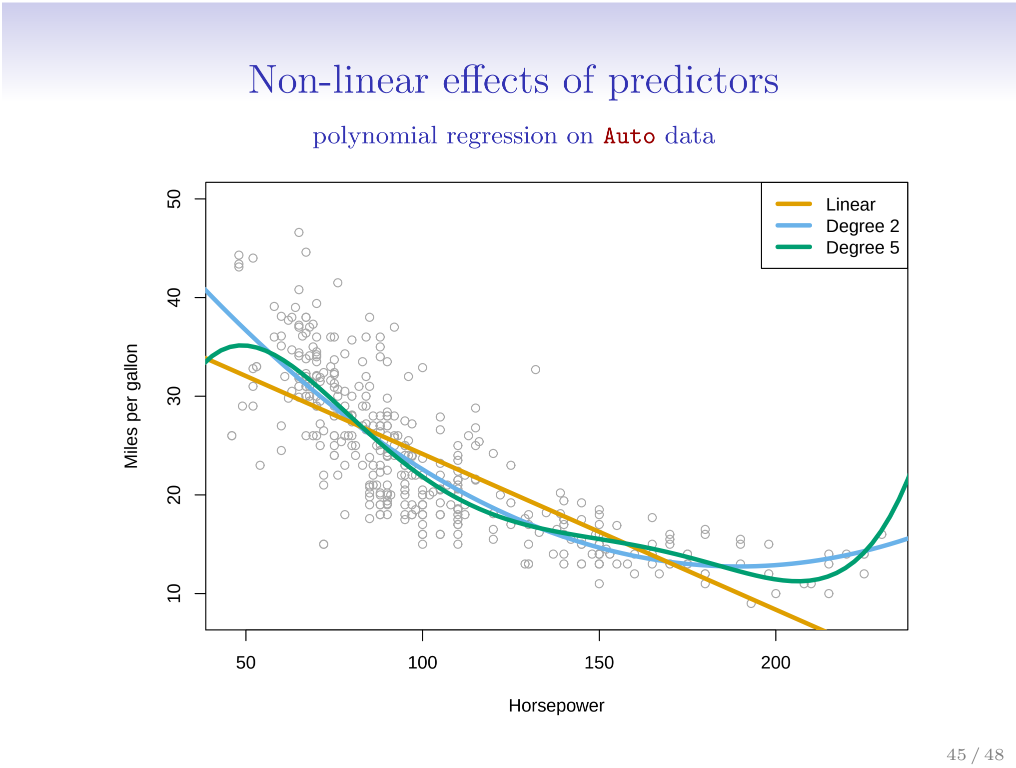

Figure 6: Polynomial regression on the Auto data. The relationship between miles per gallon and horsepower is clearly nonlinear, and polynomial terms can improve the fit.

A key lesson is that linear regression is not limited to fitting straight lines in the original variables. By introducing transformed predictors, we can fit a rich variety of nonlinear relationships while still using the familiar least squares machinery.

Of course, once we allow polynomial terms, an immediate question arises: what degree polynomial should we use? A quadratic model may not be flexible enough, while a very high-degree polynomial may overfit the data.

For example, we could consider the family of models

$$Y = \beta_0 + \beta_1 X + \cdots + \beta_d X^d + \varepsilon,$$

for degrees

$$d = 1, 2, 3, 4, 5, \ldots.$$

Each choice of d corresponds to a different linear model. *The problem of deciding which model to use is known as* ***model selection****.* One way to solve this problem is to use **Cross-Validation**, which we will discuss later in the course.

The figure suggests that

$$\texttt{mpg} = \beta_0 + \beta_1 \times \texttt{horsepower} + \beta_2 \times \texttt{horsepower}^2 + \epsilon$$

may provide a better fit.

	Coefficient	Std. Error	t-statistic	p-value
Intercept	56.9001	1.8004	31.6	< 0.0001
horsepower	-0.4662	0.0311	-15.0	< 0.0001
$\texttt{horsepower}^2$	0.0012	0.0001	10.1	< 0.0001

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Figure 7: Polynomial fits of increasing degree. As the degree increases, the fitted curve becomes more flexible. The challenge is to find a model that captures the signal without fitting the noise.

Model selection is one of the central problems in statistics and machine learning. A model that is too simple may fail to capture important patterns in the data, while a model that is too complex may fit random noise and perform poorly on new observations.

We will return to this topic throughout the course.

1.6.2 Interactions

An additive linear model assumes that the effect of one predictor does not depend on the value of another predictor. For example,

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \varepsilon$$

assumes that the effect of TV advertising is the same regardless of the amount spent on radio advertising.

But in many applications, predictors interact. In the advertising example, TV and radio may have a synergy effect. Spending on both together, that is if we have a budget of 100, distributing it equally may be more effective than spending on either alone.

One common way to encode interaction in a linear model is

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot (\text{TV} \times \text{radio}) + \varepsilon.$$

This can be rewritten as

$$\text{sales} = \beta_0 + (\beta_1 + \beta_3 \cdot \text{radio}) \cdot \text{TV} + \beta_2 \cdot \text{radio} + \varepsilon.$$

Thus the effect of TV depends on the value of radio. In effect, we now have the coefficient of TV as a linear function of radio; equivalently, we have introduced an extra predictor which is the product of radio and TV. This is analogous to introducing extra predictors $X^2, X^3, \dots$ in a polynomial regression model.

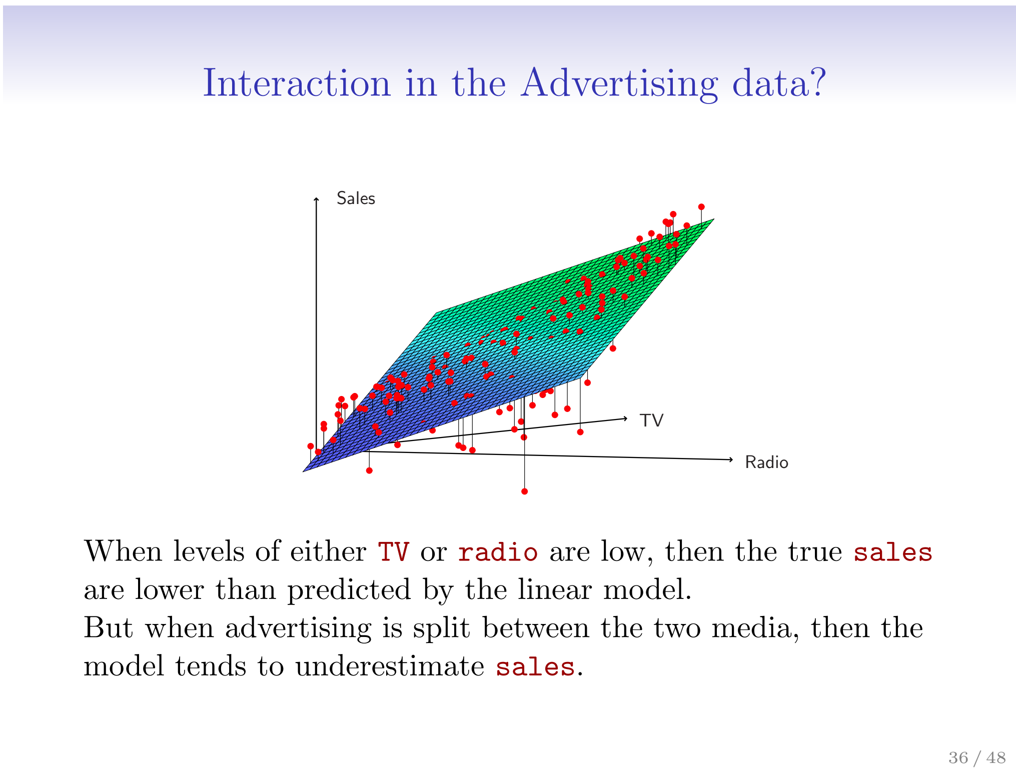

Figure 8: Interaction in the advertising data. The additive linear model misses a synergy effect between TV and radio advertising.

2 Probabilistic/Statistical Inference

Okay so we have a method to fit linear functions to data. How good is this method? Does it even make sense to use this method? I guess we can start by making some assumptions, such as the linear model is true. This is a rather strong assumption but is a starting point. So much has been done under this assumption which may not even hold! Putting on our pedantic hat again, once we make this assumption, now we can ask how close is the fitted straight line/plane to the true line/plane probabilistically? This question opens pandora's boxes and falls within the realm of statistical inference.

2.1 Frequentist Inference for Linear Regression

So far we have focused on fitting a linear regression model. Given data

$$(y_i, x_{i1}, \dots, x_{im}), \qquad i = 1, \dots, n,$$

we compute the least squares estimator

$$\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)^T.$$

A natural question is: how much should we trust these estimated coefficients?

For example, suppose the fitted coefficient of TV advertising is

$$\hat{\beta}_{TV} = 0.045.$$

Is this evidence that TV advertising truly affects sales, or could such a coefficient arise simply because of random variation in the data?

To answer such questions, we need a framework for quantifying uncertainty.

2.1.1 The Frequentist Viewpoint

The key idea in frequentist statistics is that the observed dataset is only one realization of a random experiment.

Imagine that nature generates data according to the linear model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_m X_{im} + \varepsilon_i,$$

where

$$\varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

If we repeatedly collected new datasets from the same population and recomputed the least squares estimator each time, we would obtain different values of

$$\hat{\beta}.$$

Thus the least squares estimator should itself be viewed as a random variable.

The goal of frequentist inference is to understand the distribution of this random variable. Once we know its distribution, we can construct confidence intervals and perform hypothesis tests.

2.1.2 Distribution of the Least Squares Estimator.

Using matrix notation, write

$$y = X\beta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \sigma^2 I_n).$$

The least squares estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Substituting

$$y = X\beta + \varepsilon$$

gives

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= \beta + (X^T X)^{-1} X^T \varepsilon.\end{aligned}$$

Now we can go two ways. We can assume that the rows of X are also random or assume that X is fixed. The dependence of $\hat{\beta}$ on X is complicated. We will assume that the design matrix X is fixed, the only randomness comes from ε.

Since ε is multivariate normal, it follows that $\hat{\beta}$ is also multivariate normal:

$$\hat{\beta} \sim N\big(\beta, \sigma^2 (X^T X)^{-1}\big).$$

This formula is one of the most important results in linear regression. It tells us that the least squares estimator is centered at the true coefficient vector β, and it quantifies the variability of the estimator through the covariance matrix

$$\sigma^2 (X^T X)^{-1}.$$

Looking at the j-th component, we obtain

$$\hat{\beta}_j \sim N\Big(\beta_j, \sigma^2 (X^T X)^{-1}_{j+1,j+1}\Big),$$

where $(X^T X)^{-1}_{j+1,j+1}$ denotes the appropriate diagonal entry of $(X^T X)^{-1}$.

Consequently,

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^T X)^{-1}_{j+1,j+1}}} \sim N(0,1).$$

2.1.3 Estimating the Noise Level.

The previous result assumes that the noise standard deviation σ is known. In practice, σ is almost never known and must be estimated from the data.

The residual sum of squares is

$$RSS = \|y - X\hat{\beta}\|^2. \tag{2}$$

The standard estimate of σ is the residual standard error

$$\hat{\sigma} = \sqrt{\frac{RSS}{n-m-1}}.$$

After replacing σ by $\hat{\sigma}$, the standard normal distribution is replaced by a Student t-distribution:

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{(X^T X)^{-1}_{j+1,j+1}}} \sim t_{n-m-1}. \tag{3}$$

Exercise: Justify the last two displays.

2.1.4 Confidence Intervals.

Let $t_{n-m-1,\alpha/2}$ denote the upper $\alpha/2$-quantile of the t-distribution with $n-m-1$ degrees of freedom.

Then a $100(1-\alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-m-1,\alpha/2}\hat{\sigma}\sqrt{(X^T X)^{-1}_{j+1,j+1}}.$$

The quantity

$$\hat{\sigma}\sqrt{(X^T X)^{-1}_{j+1,j+1}}$$

is called the standard error of $\hat{\beta}_j$.

A confidence interval is often misinterpreted as saying that β_j has probability $1-\alpha$ of lying inside the interval. This is not the frequentist interpretation.

In the frequentist framework, the parameter β_j is fixed and unknown, while the interval is random because it depends on the random data. If we repeatedly generated new datasets and constructed a confidence interval each time, then approximately $100(1-\alpha)\%$ of those intervals would contain the true value of β_j.

2.1.5 Reading Regression Outputs

The figure suggests that

$$\texttt{mpg} = \beta_0 + \beta_1 \times \texttt{horsepower} + \beta_2 \times \texttt{horsepower}^2 + \epsilon$$

may provide a better fit.

	Coefficient	Std. Error	t-statistic	p-value
`Intercept`	56.9001	1.8004	31.6	< 0.0001
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Figure 9: Polynomial fits of increasing degree. As the degree increases, the fitted curve becomes more flexible. The challenge is to find a model that captures the signal without fitting the noise.

How do we read the above linear regression output? The coefficients are the OLS solution $\hat{\beta}$ (presumably its well defined here). What is the *Std. error* column? In this model, we know the variance of $\hat{\beta}$ explicitly. For example, the variance of $\hat{\beta}_j$ is equal to $\sigma^2(X^TX)^{-1}_{j+1,j+1}$. This column gives the square root of this quantity where σ^2 is replaced by its estimate given in (2). So this is simply an estimate of the true standard deviation of $\hat{\beta}$. *Should we also calculate an estimate of the standard deviation of this estimate? This seems to be a never ending loop but people usually stop here.*

Now what is the *t statistic* column giving? Here one does calculations assuming $\beta_j = 0$. This is supposedly the null hypothesis asserting that the jth predictor is not significant. This column then simply computes the quantity given in (3) setting $\beta_j = 0$.

The *p-value* is then given by the probability (assuming $\beta_j = 0$) of a random variable distributed as t with $n - m - 1$ degrees of freedom exceeding the observed t statistic in magnitude. Often one looks at this pvalue, sees if it is less than 0.05 and if so, concludes that this predictor is *significant*.

Remark 1. *A small p-value does not imply that the actual effect is large or practically important. With sufficiently large sample sizes, even very small effects can produce highly significant p-values.*

Remark 2. *These frequentist calculations rely heavily on the normality assumption of the errors.*

In case we assume a different distribution, these calculations are not valid for any sample size n. But it is plausible that under mild assumptions on the covariates, the distributional calculations can be valid asymptotically. We will need (some variant of) central limit theorem to show this formally.

2.2 Bayesian Inference for Linear Regression

The frequentist approach views the coefficient vector

$$\beta = (\beta_0, \beta_1, \ldots, \beta_m)^T$$

as an unknown but fixed quantity.

The Bayesian approach takes a different point of view. Here the coefficients are treated as random variables. Before seeing the data, we express our uncertainty about the coefficients through a prior distribution. After observing the data, we update this prior using Bayes' rule and obtain a posterior distribution.

The posterior distribution contains all information about the unknown coefficients after observing the data.

We again assume the standard Gaussian linear model

$$y = X\beta + \varepsilon,$$

with

$$\varepsilon \sim N(0, \sigma^2 I_n).$$

2.2.1 A Default Prior.

A common default Bayesian choice is to place a very diffuse prior on all regression coefficients and on the logarithm of the noise level.

Formally, we may imagine

$$\beta_0, \beta_1, \ldots, \beta_m, \log\sigma \overset{\text{ind}}{\sim} \text{Uniform}(-C, C),$$

where C is very large.

This prior expresses the idea that, before seeing the data, we have little information about the coefficients or the noise level.

2.2.2 Posterior Distribution

Under the Gaussian model, the likelihood is

$$p(y \mid \beta, \sigma) \propto \sigma^{-n} \exp\left(-\frac{1}{2\sigma^2}\|y - X\beta\|^2\right).$$

The quantity

$$S(\beta) = \|y - X\beta\|^2$$

is the residual sum of squares.

Combining the likelihood with the diffuse prior gives the joint posterior density

$$p(\beta, \sigma \mid y) \propto \sigma^{-n} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) \cdot \frac{1}{\sigma},$$

where the extra factor $1/\sigma$ comes from using a flat prior on $\log \sigma$. Therefore

$$p(\beta, \sigma \mid y) \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right).$$

To obtain the marginal posterior density of β, we integrate out σ:

$$p(\beta \mid y) \propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma.$$

Let

$$u = \frac{\sqrt{S(\beta)}}{\sigma}.$$

Then

$$\sigma = \frac{\sqrt{S(\beta)}}{u}$$

and

$$d\sigma = -\sqrt{S(\beta)} u^{-2} \, du.$$

Thus

$$\begin{aligned}
p(\beta \mid y) &\propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma \\
&= \int_\infty^0 \left(\frac{\sqrt{S(\beta)}}{u}\right)^{-n-1} \exp\left(-\frac{u^2}{2}\right) \left(-\sqrt{S(\beta)} u^{-2}\right) du \\
&= (S(\beta))^{-n/2} \int_0^\infty u^{n-1} \exp\left(-\frac{u^2}{2}\right) du.
\end{aligned}$$

The integral

$$\int_0^\infty u^{n-1} \exp\left(-\frac{u^2}{2}\right) du$$

does not depend on β. Therefore, as a function of β,

$$p(\beta \mid y) \propto (S(\beta))^{-n/2}.$$

Remark 3. *The prior density is a decreasing function of $S(\beta)$ which makes sense. So the posterior mode is at the OLS estimate.*

Using the Pythoagorean identity (1),

$$S(\beta) = S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X(\beta - \hat{\beta}).$$

Substituting this into the posterior density gives

$$p(\beta \mid y) \propto \left[\frac{1}{1 + (\beta - \hat{\beta})^T \frac{X^T X}{S(\hat{\beta})}(\beta - \hat{\beta})} \right]^{n/2}. \tag{4}$$

It turns out that this is exactly the density of a multivariate Student t-distribution.

2.3 The Multivariate Student t-Distribution

It is useful to briefly review the multivariate t distribution.

Definition. A random vector $T \in \mathbb{R}^p$ is said to have a multivariate t-distribution with location vector μ, scale matrix Σ, and ν degrees of freedom if its density is

$$f(t) = C\left(1 + \frac{1}{\nu}(t-\mu)^T \Sigma^{-1}(t-\mu)\right)^{-(\nu+p)/2},$$

where C is a normalizing constant chosen so that the density integrates to one.
We write

$$T \sim t_p(\mu, \Sigma, \nu).$$

The formula above should look familiar. Recall that if

$$X \sim N_p(\mu, \Sigma),$$

then the density of X is proportional to

$$\exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right).$$

Thus both the normal and t-distributions involve exactly the same quadratic form

$$(x-\mu)^T \Sigma^{-1}(x-\mu).$$

The difference is that the normal distribution has exponential tails, whereas the t-distribution has polynomial tails. Consequently, the t-distribution places more probability on extreme values.

The multivariate t-distribution can be obtained from a multivariate normal distribution.

Suppose

$$X \sim N_p(\mu, \Sigma)$$

and

$$V \sim \chi^2_\nu$$

are independent.

Define

$$T = \mu + \frac{X - \mu}{\sqrt{V/\nu}}.$$

Then

$$T \sim t_p(\mu, \Sigma, \nu).$$

One of the most useful properties of the multivariate t-distribution is that every linear combination of its coordinates is again t-distributed.

If

$$T \sim t_p(\mu, \Sigma, \nu)$$

and

$$a \in \mathbb{R}^p,$$

then

$$a^T T \sim t_1(a^T \mu, a^T \Sigma a, \nu).$$

In particular, each coordinate T_j satisfies

$$T_j \sim t_1(\mu_j, \Sigma_{jj}, \nu).$$

This fact will be extremely important later because it allows us to derive the posterior distribution of each regression coefficient β_j from the joint posterior distribution of the entire vector β.

2.3.1 Large Degrees of Freedom

When the degrees of freedom ν are large, the multivariate t-distribution becomes very close to a multivariate normal distribution.

To understand why, consider the density

$$f(t) \propto \left(1 + \frac{1}{\nu}(t-\mu)^T\Sigma^{-1}(t-\mu)\right)^{-(\nu+p)/2}.$$

When ν is large, the quantity

$$\frac{1}{\nu}(t-\mu)^T\Sigma^{-1}(t-\mu)$$

is small for values of t near the center of the distribution. Using the approximation

$$1 + z \approx e^z$$

for small z, we obtain

$$\left(1 + \frac{1}{\nu}(t-\mu)^T\Sigma^{-1}(t-\mu)\right)^{-(\nu+p)/2} \approx \exp\left(-\frac{\nu+p}{2\nu}(t-\mu)^T\Sigma^{-1}(t-\mu)\right).$$

Since

$$\frac{\nu+p}{\nu} = 1 + \frac{p}{\nu} \approx 1$$

when ν is large, the density becomes approximately

$$\exp\left(-\frac{1}{2}(t-\mu)^T\Sigma^{-1}(t-\mu)\right),$$

which is precisely the kernel of the multivariate normal density

$$N_p(\mu, \Sigma).$$

Another way to see this is through the representation

$$T = \mu + \frac{X-\mu}{\sqrt{V/\nu}},$$

where

$$X \sim N_p(\mu, \Sigma), \qquad V \sim \chi^2_\nu.$$

By the Law of Large Numbers,

$$\frac{V}{\nu} \to 1$$

when ν becomes large. Thus

$$\sqrt{\frac{V}{\nu}} \approx 1,$$

and consequently

$$T \approx X.$$

Since X is multivariate normal, this again suggests that

$$t_p(\mu, \Sigma, \nu) \approx N_p(\mu, \Sigma) \tag{5}$$

for large ν.

2.3.2 Posterior Distribution of the Coefficients.

In light of the above definition of the multivariate t, by looking at (4), the posterior distribution is

$$\beta \mid y \sim t_{m+1}\left(\hat{\beta}, \frac{S(\hat{\beta})}{n-m-1}(X^T X)^{-1}, n-m-1\right).$$

Define

$$\hat{\sigma} = \sqrt{\frac{S(\hat{\beta})}{n-m-1}}.$$

This quantity is exactly the residual standard error from classical linear regression.

With this notation,

$$\beta \mid y \sim t_{m+1}\left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}, n-m-1\right).$$

Thus the posterior is centered at the least squares estimate and its spread is determined by the same covariance structure that appears in frequentist inference.

An important property of the multivariate t-distribution is that every component is again t-distributed.

Therefore

$$\beta_j \mid y \sim t_1\left(\hat{\beta}_j, \hat{\sigma}^2 (X^T X)^{-1}_{j+1,j+1}, n-m-1\right).$$

Equivalently,

$$\left.\frac{\beta_j - \hat{\beta}_j}{\hat{\sigma}\sqrt{(X^T X)^{-1}_{j+1,j+1}}}\right| y \sim t_{n-m-1}.$$

2.3.3 Bayesian Credible Intervals.

Let $t_{n-m-1,\alpha/2}$ denote the upper $\alpha/2$-quantile of the t-distribution.

Then

$$P\left(\hat{\beta}_j - t_{n-m-1,\alpha/2}\hat{\sigma}\sqrt{(X^TX)^{-1}_{j+1,j+1}} \le \beta_j \le \hat{\beta}_j + t_{n-m-1,\alpha/2}\hat{\sigma}\sqrt{(X^TX)^{-1}_{j+1,j+1}} \mid y\right) = 1-\alpha.$$

This interval is called a $100(1-\alpha)\%$ Bayesian credible interval.

Notice the interpretation. After observing the data, the parameter β_j is treated as random. The credible interval therefore has the direct probabilistic interpretation

Given the observed data, there is probability $1-\alpha$ that β_j lies in the interval.

This is fundamentally different from the frequentist interpretation of a confidence interval. No repeated draws from the model is necessary for this interpretation.

2.3.4 A Coincidence.

For the diffuse prior used above, the Bayesian credible interval turns out to be *numerically identical* to the classical frequentist confidence interval.

The calculations are different and the interpretations are different, but the final interval is exactly the same.

This coincidence is special to this particular model and prior choice. In general settings, Bayesian and frequentist intervals need not agree.

2.3.5 Large Sample Behavior.

The posterior distribution above is a multivariate t-distribution with $n-m-1$ degrees of freedom.

When $n-m-1$ is large, the t-distribution is very close to a normal distribution by (5). Therefore

$$\beta \mid y \approx N\left(\hat{\beta}, \hat{\sigma}^2(X^TX)^{-1}\right).$$

Similarly,

$$\beta_j \mid y \approx N\left(\hat{\beta}_j, \hat{\sigma}^2(X^TX)^{-1}_{j+1,j+1}\right).$$

Thus for large sample sizes, Bayesian uncertainty quantification for linear regression becomes approximately Gaussian.

2.4 Autoregression in Time Series

We now consider an example which is useful to illustrate the differences between frequentist and bayesian inference. Suppose we observe a time series

$$y_1, \ldots, y_n.$$

The AR(1) model is

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t, \qquad t = 2, \ldots, n.$$

Question:When would such a model be appropriate?

We will treat y_1 as deterministic. This is a linear model with response y_t and predictor y_{t-1}.

Define

$$y = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix},$$

and

$$X = \begin{pmatrix} 1 & y_1 \\ 1 & y_2 \\ \vdots & \vdots \\ 1 & y_{n-1} \end{pmatrix}.$$

Then

$$y = X\beta + \varepsilon,$$

where

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The least squares estimator still has the formal expression

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

However, here X contains

$$y_1, \ldots, y_{n-1},$$

which are part of the observed time series. Therefore X is random and dependent on the response vector. In particular, we can no longer use the fact that $\hat{\beta}$ being a linear function of y is also normally distributed.

For example,

$$\hat{\beta}_1 = \frac{\sum_{t=2}^n (y_t - \bar{y}_{2:n})(y_{t-1} - \bar{y}_{1:n-1})}{\sum_{t=2}^n (y_{t-1} - \bar{y}_{1:n-1})^2},$$

where

$$\bar{y}_{2:n} = \frac{y_2 + \cdots + y_n}{n-1},$$

and

$$\bar{y}_{1:n-1} = \frac{y_1 + \cdots + y_{n-1}}{n-1}.$$

The exact finite-sample distribution of $\hat{\beta}_1$ is generally quite complicated because the covariates are themselves random and dependent on the response process. Nevertheless, asymptotic arguments can be used to study the behavior of $\hat{\beta}_1$ when the sample size becomes large.

Under suitable regularity conditions, one can show that

$$\sqrt{n}\,(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, \tau^2),$$

for an appropriate asymptotic variance τ^2. The proof relies on a central limit theorem for dependent random variables, reflecting the fact that the observations in a time series are not independent; For details, see Shumway and Stoffer [Shumway and Stoffer(2017)].

For Bayesian inference, we again place the prior

$$\beta_0, \beta_1, \log\sigma \propto 1.$$

By the chain rule,

$$p(y_1, \ldots, y_n \mid \beta, \sigma) = p(y_1 \mid \beta, \sigma) \prod_{t=2}^{n} p(y_t \mid y_1, \ldots, y_{t-1}, \beta, \sigma).$$

Assume

$$\varepsilon_t \sim N(0, \sigma^2),$$

and that ε_t is independent of the past.

Then

$$y_t \mid y_1, \ldots, y_{t-1}, \beta, \sigma \sim N(\beta_0 + \beta_1 y_{t-1}, \sigma^2).$$

If $p(y_1 \mid \beta, \sigma)$ does not depend on β, σ, then

$$L(\beta, \sigma) \propto \sigma^{-(n-1)} \exp\left[-\frac{1}{2\sigma^2} \sum_{t=2}^{n} (y_t - \beta_0 - \beta_1 y_{t-1})^2\right].$$

In matrix notation,

$$L(\beta, \sigma) \propto \sigma^{-(n-1)} \exp\left[-\frac{1}{2\sigma^2} \|y - X\beta\|^2\right].$$

This has exactly the same form as ordinary linear regression, except that the sample size is $n-1$. Since the prior is also the same, we can use the exact same reasoning to derive the posterior distribution of β as a multivariate t.

Since there are two regression coefficients, the degrees of freedom are

$$(n-1)-2=n-3.$$

Therefore

$$\beta \mid y \sim t_2\left(\hat{\beta}, \frac{S(\hat{\beta})}{n-3}(X^\top X)^{-1}, n-3\right).$$

Equivalently, with

$$\hat{\sigma}^2 = \frac{S(\hat{\beta})}{n-3},$$

we have

$$\beta \mid y \sim t_2\left(\hat{\beta}, \hat{\sigma}^2(X^\top X)^{-1}, n-3\right).$$

Thus Bayesian inference for the AR(1) model gives a posterior t-distribution by essentially the same calculation as ordinary linear regression, while frequentist inference requires a more delicate analysis of the sampling distribution.

Remark 4. *The above is an example where the X matrix is random. In that case, to derive the sampling distribution of $\hat{\beta}$ as a frequentist, one needs to take the randomness of both X, y into account. For a Bayesian, the fact that X and y is random does not really matter once the likelihood and the prior has been decided on.*

Unlike in the frequentist approach, the Bayesian machinery does not look at other hypothetical data sets or repeated draws from the model. For the Bayesian, the data which is observed is fixed, and the only thing that matters is the form of the prior and the likelihood as a function of the unknown parameters. This means we do not have to think about a hard probability question such as what is the sampling distribution of the least squares coefficients.

References

[Shumway and Stoffer(2017)] Robert H. Shumway and David S. Stoffer. *Time Series Analysis and Its Applications: With R Examples.* Springer Texts in Statistics. Springer, Cham, Switzerland, 4 edition, 2017. doi: 10.1007/978-3-319-52452-8.